

KisanSetu

Smart Procurement & Direct Market Access Platform for Farmers

SIH 2026 Integrated Solution

Combining SIH26032 — Farmer procurement status & waiting-time uncertainty
with SIH26033 — Middlemen, farmer income & consumer price transparency

Project Documentation

A complete concept, feature, architecture, implementation and demonstration document for the proposed unified agricultural platform.

1. Executive Summary

KisanSetu is a unified digital platform designed to address two connected problems in agricultural procurement and market access: uncertainty around waiting at procurement centers and limited transparency when farmers sell their produce.

The platform combines **smart procurement management**, **live queue visibility**, **slot booking**, **waiting-time estimation**, **direct buyer discovery**, **price comparison** and **logistics matching** into one farmer-centric system.

The central idea is not to build two separate applications. It is to connect the farmer's journey from preparing produce for procurement through finding buyers and arranging transportation. The result is a single digital workflow that improves information availability, reduces avoidable waiting and gives farmers more visibility into market opportunities.

2. Problem Statements

SIH26032 — Farmer Procurement Status & Waiting-Time Uncertainty

Farmers visiting procurement centers may have limited visibility into queue length, current token, expected processing time, available slots and center capacity. A farmer can therefore spend hours waiting, or travel to an overloaded center without knowing its current situation.

SIH26033 — Middlemen Reducing Farmer Realization & Increasing Consumer Prices

Agricultural produce can move through several intermediaries before reaching the final buyer. Farmers may have limited information about alternative buyers and prevailing prices, which can weaken bargaining power and make the final price chain difficult to understand.

These problems are related: a farmer needs reliable information not only about *where and when to deliver*, but also about *where and how to sell*. KisanSetu addresses both information gaps through one platform.

3. Existing System & Pain Points

Procurement journey

Farmer → Travel to center → Discover queue → Wait → Procurement

- Queue information may not be available before arrival.
- Farmers may spend significant time waiting at busy centers.
- Center operators have difficulty communicating changing queue conditions to every farmer.
- Low-connectivity users may be excluded from app-only updates.

Market journey

Farmer → Local trader/intermediary → Wholesaler → Retailer → Consumer

- Farmers may not easily compare buyer offers.
- Price information can be fragmented across locations and channels.
- Transport is often treated as a separate problem rather than part of the transaction.
- A farmer may lack a unified digital record of offers, orders and completed sales.

4. Proposed Solution — KisanSetu

KisanSetu provides four integrated layers:

- **Smart Procurement:** center discovery, slot booking, digital tokens, live queue and estimated waiting time.
- **Direct Marketplace:** produce listings, buyer discovery, offers and order management.
- **Price Transparency:** reference prices and side-by-side buyer offers to improve decision-making.
- **Logistics Matching:** connect accepted trades with suitable transportation based on distance and capacity.

The platform also includes an **SMS/IVR fallback** so that important procurement updates do not depend entirely on continuous smartphone internet access.

5. End-to-End Farmer Workflow

1. **Register:** create a farmer profile with basic contact and location information.
2. **Add produce:** specify crop, quantity, availability and expected price.
3. **Find procurement center:** compare centers using distance, queue and estimated waiting time.
4. **Book slot:** select an available procurement window and receive a digital token.
5. **Track queue:** monitor current token, farmers ahead and estimated waiting time.
6. **Receive notification:** get app, SMS or IVR updates as the turn approaches.
7. **Explore market:** compare reference prices and available buyer offers.
8. **Select buyer:** accept or negotiate an offer.
9. **Match logistics:** find a suitable vehicle/transport option.
10. **Complete transaction:** maintain a digital order and transaction record.

6. Smart Procurement Module

The procurement module turns an uncertain visit into a planned visit.

Feature	Purpose
Center discovery	Compare nearby procurement centers.
Slot booking	Distribute arrivals across available time windows.
Digital token	Give each farmer a clear position in the queue.
Live queue	Show current serving token and farmer position.
Waiting-time estimation	Estimate remaining time from queue and processing data.
Capacity monitoring	Help operators identify overloaded centers.
Notifications	Inform farmers when their slot/turn is approaching.
SMS/IVR fallback	Support farmers with limited connectivity.

7. Waiting-Time Estimation

The first prototype can use a transparent calculation rather than an unexplained AI prediction.

Basic estimation:

Estimated Waiting Time $\approx$ Farmers Ahead $\times$ Average Processing Time $\div$ Active Counters

The model can later incorporate historical throughput, current center load, crop type, quantity and observed processing speed. The system should clearly label the value as an estimate rather than a guarantee.

8. Direct Marketplace Module

Farmers can create produce listings containing crop, quantity, location, availability and expected price. Buyers can search by crop, quantity and distance.

Actor	Key capabilities
Farmer	List produce, view prices, receive offers, negotiate, accept/reject offers, track orders.
Buyer	Search produce, discover farmers, compare offers, make offers, place orders.
Platform	Match produce with buyers, calculate logistics options and maintain transaction status.

The prototype should be scoped to one crop and one region if necessary. A focused marketplace is easier to demonstrate convincingly than a large but shallow multi-crop platform.

9. Price Transparency

The platform should help farmers compare available information rather than simply display a single price.

Example: Reference price: ₹2,500/quintal; Buyer A: ₹2,550; Buyer B: ₹2,620; Buyer C: ₹2,580.

The system can present the highest offer, distance, quantity and estimated logistics cost together. This supports better decision-making without making an unsupported claim that every transaction will produce a fixed income increase.

10. Logistics Matching

After a farmer accepts a buyer offer, KisanSetu can connect the transaction with transportation.

- Match vehicle capacity with order quantity.
- Calculate approximate route distance.
- Estimate transport cost.
- Show pickup and drop locations.
- Track logistics status in the order workflow.

This connects three parties that are often handled separately: **farmer + buyer + transporter**.

11. User Roles & Dashboards

Role	Dashboard
Farmer	My Produce, Procurement Slots, Live Queue, Market Prices, Buyer Offers, Orders, Payments.
Buyer	Find Produce, Nearby Farmers, Price Comparison, Offers, Orders, Logistics.
Procurement Officer	Live Queues, Center Capacity, Counters, Today's Arrivals, Completed Procurement, Pending Farmers.
System Admin	User management, center configuration, marketplace configuration, analytics and system monitoring.

12. System Architecture

Client layer

Farmer Web/App • Buyer Portal • Admin Dashboard

Application layer

API Gateway • Authentication • Procurement Service • Marketplace Service • Logistics Service • Notification Service

Core engines

Slot Engine • Queue Engine • Waiting-Time Estimator • Offer Engine • Price Comparison • Matching Engine

Communication layer

Push Notifications • SMS • IVR

Data layer

PostgreSQL database and supporting cache/realtime infrastructure.

The architecture is modular so procurement and marketplace functionality can evolve independently while sharing farmer identity, produce data, notifications and transaction information.

13. Suggested Technology Stack

Layer	Technology
Frontend	React / Next.js
Backend	Node.js + Express or Python + FastAPI
Database	PostgreSQL
Realtime queue	WebSockets / Socket.IO
Authentication	OTP + JWT / role-based access
Maps	Google Maps API or OpenStreetMap-based solution
Notifications	SMS, IVR and push notification provider
Deployment	Cloud hosting with managed database

14. Core Data Model

Entity	Important fields
Farmer	farmer_id, name, contact, location
Produce	produce_id, farmer_id, crop, quantity, expected_price, availability
Procurement Center	center_id, location, capacity, active_counters
Procurement Slot	slot_id, center_id, farmer_id, date, time, token, status
Queue	center_id, token, position, estimated_wait, status
Buyer Offer	offer_id, produce_id, buyer_id, price, quantity, status
Order	order_id, farmer_id, buyer_id, produce_id, quantity, price, status
Logistics	vehicle_id, capacity, route, cost, status

15. Security & Trust

- Role-based authorization for farmers, buyers, officers and administrators.
- Secure authentication and encrypted communication.
- Input validation and API protection.
- Minimal collection of personal information.
- Audit trail for important procurement and transaction events.
- Clear distinction between estimated values and confirmed transaction values.

16. What Makes KisanSetu Different

The differentiation is the **integration** rather than any single isolated feature.

Typical approach	KisanSetu
Queue management only	Queue + slot + waiting estimate + notifications
Marketplace only	Marketplace + procurement workflow
Price display	Reference information + multiple buyer offers
Marketplace without logistics	Buyer matching + transport matching
App-only access	App/Web + SMS/IVR fallback
Separate workflows	One connected farmer journey

17. Innovation

- **Smart waiting-time estimation:** convert queue position into an understandable time estimate.
- **Procurement + marketplace integration:** connect physical procurement operations with digital market access.
- **Multi-channel access:** support smartphone users and low-connectivity users.
- **Decision-oriented price transparency:** compare offers with quantity, distance and logistics context.
- **End-to-end transaction flow:** farmer → produce → buyer → logistics → order.

18. Expected Impact

Stakeholder	Expected benefit
Farmers	Better visibility, reduced avoidable waiting, more buyer choices and improved access to price information.
Procurement centers	Better queue planning, capacity visibility and smoother farmer arrivals.
Buyers	Easier discovery of local produce and more structured purchasing.
Transporters	More discoverable demand and better vehicle utilization.
System operators	Centralized operational data and analytics.

For the hackathon, impact should be presented as target outcomes rather than unsupported percentage claims.

19. MVP Scope for SIH

To keep the implementation realistic, the first working prototype should focus on a controlled scenario.

- One crop and one selected region.
- 3–5 simulated procurement centers.
- Farmer registration and produce listing.
- Center comparison and slot booking.
- Real-time simulated queue updates.
- Waiting-time estimation.
- Buyer listings and offers.
- Basic price comparison.
- Basic logistics matching.
- SMS/IVR simulation or one working notification channel.
- Admin dashboard.

The prototype can use mock APIs and seeded data. The architecture should be designed so real government/market integrations can be connected later when authorized APIs become available.

20. Demo Scenario

Scenario: Ramesh has 800 kg of wheat.

He opens KisanSetu and compares three procurement centers. Center A has a long queue, Center B has a shorter queue, and Center C is close but heavily loaded. He chooses Center B and books a slot.

The system generates a digital token and shows the current serving token, farmers ahead and estimated waiting time. As the queue moves, the dashboard updates in real time. When his turn approaches, he receives a notification.

Ramesh also lists his wheat on the marketplace. Three buyers send offers. He compares price, quantity and approximate distance, accepts the most suitable offer, and the platform identifies a transport option.

This single story demonstrates both problem statements without making the project feel like two separate products.

21. Demo Screen Sequence

- Landing / Login
- Farmer Dashboard
- Add Produce
- Procurement Center Comparison
- Slot Booking
- Digital Token
- Live Queue
- Waiting-Time Notification
- Marketplace Listing
- Buyer Offers / Price Comparison
- Logistics Match

- Order Confirmation
- Admin Procurement Dashboard

The demo should follow the farmer's story in this order rather than jumping randomly between features.

22. Implementation Roadmap

Phase	Work
Phase 1	UI/UX, database schema, authentication and seeded data.
Phase 2	Farmer profile, produce listing and procurement center module.
Phase 3	Slot booking, queue engine and realtime updates.
Phase 4	Marketplace, offers and price comparison.
Phase 5	Logistics matching and notifications.
Phase 6	Admin analytics, security hardening, testing and deployment.

23. Future Scalability

- Expand from one crop/region to multiple crops and districts.
- Integrate authorized procurement and market-price APIs.
- Add digital payments and verified transaction workflows.
- Introduce demand and supply forecasting.
- Improve waiting-time prediction using historical center data.
- Add multilingual interfaces and stronger IVR flows.
- Integrate third-party logistics providers.
- Provide analytics for procurement authorities and market operators.

24. Risks & Mitigation

Risk	Mitigation
No real government API access during hackathon	Use realistic mock APIs and clearly document the integration contract.
Inaccurate waiting estimate	Use transparent estimation and label it as approximate.
Marketplace trust issues	Use verified profiles, transaction records and clear order states.
Low connectivity	Provide SMS/IVR fallback for critical updates.
Too broad an MVP	Start with one crop, one region and simulated centers.
Price data quality	Clearly identify data source/type and avoid unsupported claims.

25. Key Pitch

KisanSetu connects two critical moments in a farmer's journey: getting produce procured efficiently and finding transparent market opportunities.

Instead of making farmers wait blindly at procurement centers or sell with limited market visibility, the platform provides one place to plan procurement, track the queue, compare market opportunities, connect

with buyers and arrange logistics.

Core message: From waiting blindly to knowing when to go. From selling blindly to knowing whom to sell to.

26. One-Minute Explanation for Judges

“KisanSetu is a unified agricultural platform that addresses two connected problems faced by farmers. First, farmers often lack visibility into procurement-center queues, slots and waiting times. Second, they may have limited access to transparent buyer and price information. Our platform lets a farmer register produce, compare procurement centers, book a slot, receive a digital token, track the live queue and get waiting-time notifications through app, SMS or IVR. The same produce can then be listed on our direct marketplace, where farmers can compare buyer offers and match accepted orders with logistics. The key difference is that we are not building just a queue-management system or another marketplace; we connect procurement, market access and logistics into one farmer-centric workflow.”